

Postly AI

Product Requirements Document (PRD)

AI-Powered Marketing Automation for Nigerian SMEs

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Executive Summary

What is Postly AI?

Postly AI is a web-based SaaS tool that uses artificial intelligence to write marketing content for Nigerian small businesses. Instead of paying a social media manager N50,000-N150,000/month, a business owner types one sentence about what they want to say, and Postly AI writes ready-to-post captions for Instagram, WhatsApp, Twitter/X, and SMS broadcasts in seconds.

The one-line pitch

"Postly AI writes your marketing content so you can focus on running your business."

Why now?

Nigeria has over 40 million small businesses - most cannot afford a dedicated marketer.

Social media is now the primary sales channel for Nigerian SMEs (Instagram, WhatsApp, TikTok).

AI writing tools exist but are not tailored for Nigerian context, tone, or platforms.

The Vercel AI SDK makes streaming AI responses into a web app faster and cheaper than ever.

No competitor owns this space in Nigeria today - it is wide open.

Key fact	Detail
Target market	Nigerian SMEs - fashion, food, logistics, retail, beauty, services
Primary platforms	Instagram, WhatsApp, Twitter/X, SMS
Revenue model	Monthly SaaS subscription via Paystack
MVP build time	4-6 weeks for a solo developer
Starting price	N5,000/month per business
Breakeven	~25 paying customers

The Problem

2.1 Who is the customer?

The primary customer is a Nigerian small business owner aged 25-45 who runs their business primarily through Instagram DMs and WhatsApp. They sell fashion, food, beauty products, logistics services, or professional services. They are active on social media personally but struggle to market their business consistently.

Customer profile	Details
Business type	Fashion, food/catering, beauty/hair, logistics, retail, services
Staff size	1-20 employees
Location	Lagos, Abuja, Port Harcourt, Ibadan - major Nigerian cities
Tech comfort	Uses Instagram, WhatsApp, Google - comfortable with web apps
Monthly revenue	N200,000 - N5,000,000
Marketing budget	N0 - N50,000/month

2.2 What pain are we solving?

I don't know what to post

They have a product but stare at their phone for 30 minutes trying to write a caption. They post inconsistently or not at all.

No time to market

They are busy running the business. Marketing always comes last.

Can't afford a social media manager

Good social media managers in Nigeria charge N50k-N150k/month. Most SMEs cannot justify this.

Posts don't sound professional

When they do post, the copy feels flat and does not convert followers to buyers.

WhatsApp broadcasts are painful

Writing a broadcast message that doesn't sound spammy takes skill they don't have.

The Cost of This Problem

A business that posts inconsistently loses followers, loses top-of-mind awareness, and loses sales. Businesses that post 3-5x per week on Instagram get 2-3x more enquiries than those that post once a week. For a business doing N1M/month in sales, better marketing could mean N200k-N500k extra per month. Postly AI is the cheapest way to get there.

Product Overview

3.1 What Postly AI does

Postly AI is a web app. A business owner signs up, tells the tool about their business once, then generates marketing content on demand. The AI knows their industry, their tone, their target customers, and the platform they are posting on - so every piece of content feels like it was written by a professional marketer who knows their brand.

Feature	What it does
Business profile setup	Business name, industry, what they sell, tone of voice, target audience. Done once, used forever.
Content type selection	Instagram caption, WhatsApp broadcast, Twitter post, SMS promo, product spotlight, story CTA.
One-line input	User types: '50% off shoes this weekend' - AI writes 3 variations.
AI streaming output	Content appears word by word using Vercel AI SDK streaming - feels fast and premium.
Copy & use	One click to copy. User pastes directly into Instagram, WhatsApp, or SMS tool.
Content history	All generated content is saved. User can revisit, edit, or regenerate.
Weekly content plan	AI suggests 5-7 post ideas for the week based on industry and season.

3.2 Key value proposition

Without Postly AI	With Postly AI
30 min to write one caption	Under 2 minutes
N50,000-N150,000/month for SMM	N5,000-N12,000/month
Inconsistent posting	Always have something to post
Generic, unconvincing copy	Professional, Nigerian-context copy

Forgot to send WhatsApp broadcast

Broadcasts ready in seconds

Features & User Stories

4.1 Core features - MVP (what you build first)

These are the only features needed to launch and charge money. Keep it lean.

F-01

User authentication

2 days

As a business owner, I want to create an account and log in securely so my content and settings are saved.

- Email + password signup via Supabase Auth
- Google OAuth login (optional in v1)
- Protected dashboard routes using Next.js middleware

F-02

Business profile setup

1-2 days

As a user, I want to tell the AI about my business once so every piece of content it generates already understands my brand.

- Form: business name, industry, what they sell, target customer, tone (Fun/Professional/Casual/Bold)
- Saved to Supabase on first login
- Editable from settings at any time
- Profile injected into every AI prompt

F-03

AI content generator

3-4 days

As a user, I want to type one sentence about my post and get 3 ready-to-use variations in seconds.

- Content type selector: Instagram / WhatsApp / Twitter / SMS / Product spotlight
- Single text input: 'What is this post about?'
- Optional: 'Any extra detail?' (price, deadline, location)
- Calls Vercel AI SDK streamText with full business profile context
- Streams 3 variations live - user sees them generate in real time
- Copy button per variation

F-04

Content history

1-2 days

As a user, I want to see all the content I have generated before so I can reuse or edit it.

- All generations saved to Supabase with timestamp, content type, input, output
- Simple list view - newest first
- Copy and regenerate buttons on each item
- Basic filter by content type

F-05

Weekly content planner

2 days

As a user, I want the AI to suggest 7 post ideas for my business this week so I never run out of things to post.

- One-click 'Plan my week' button on dashboard
- AI generates 7 ideas tailored to industry and current week/season
- Each idea is a brief description - user clicks to generate full content
- Aware of Nigerian calendar events (Valentine's, Eid, Christmas, Black Friday)

F-06

Subscription & billing

3-4 days

As a user, I want to pay for my plan easily and have my account automatically upgrade or restrict based on my subscription.

- Paystack subscription integration (recurring monthly billing)
- Three plans: Starter, Growth, Pro
- Free trial: 5 generations before paywall
- Generation counter tracked in Supabase
- Plan limits enforced server-side
- Paystack webhook updates subscription status in DB

4.2 Phase 2 features (after first paying customers)

Do NOT build these for MVP. Build them only after you have 10+ paying customers.

Feature	Description
Canva integration	AI picks a template and pre-fills the caption. User downloads a ready-made post image.
Post scheduling	Schedule posts to publish directly to Instagram/Twitter at a set time.
WhatsApp broadcast sender	Upload customer list, AI writes the message, sends via WhatsApp Business API.
Analytics dashboard	See which content types the user generates most. Upsell opportunity.

Multi-business support	Manage 2+ business profiles under one account. Target agencies.
AI hashtag generator	Auto-suggest relevant Nigerian and industry-specific hashtags.
Email marketing	AI writes email newsletters for businesses with a customer email list.

Technical Architecture

5.1 Tech stack

Layer	Technology
Frontend framework	Next.js 14+ (App Router) - you already know this
Styling	Tailwind CSS - you already know this
AI integration	Vercel AI SDK (ai package) - what you are learning now
AI model	OpenAI GPT-4o-mini via @ai-sdk/openai - cheapest, fastest, excellent quality
Database	Supabase (PostgreSQL) - you already use this
Authentication	Supabase Auth - built-in, free
Payments	Paystack - Nigerian payments, subscription support
Hosting	Vercel - free tier, deploy from GitHub, zero config
Email	Resend - free tier for transactional emails

5.2 Vercel AI SDK integration

The Vercel AI SDK is the core of the product. It handles streaming AI responses to the browser, making generation feel fast and live. Here is exactly how it works:

Terminal - install dependencies

```
npm install ai @ai-sdk/openai
```

app/api/generate/route.ts - the AI route

```
import { streamText } from 'ai'
import { openai } from '@ai-sdk/openai'

export async function POST(req: Request) {
  const { contentType, userInput, extraDetail, profile } = await req.json()

  const systemPrompt = `You are a professional Nigerian marketing copywriter.
  Business: ${profile.businessName}`
```

```

Industry: ${profile.industry}
They sell: ${profile.whatTheySell}
Target customer: ${profile.targetCustomer}
Tone: ${profile.tone}
Always write in a way that feels local, relatable, and drives action.`

    const userPrompt = `Write 3 variations of a ${contentType} for: "${userInput}"
${extraDetail ? 'Extra detail: ' + extraDetail : ''}
Number each variation (1. 2. 3.).
Keep each under 150 words. Include a call to action in each.`

    const result = streamText({
      model: openai('gpt-4o-mini'),
      system: systemPrompt,
      prompt: userPrompt,
      maxTokens: 600,
    })

    return result.toDataStreamResponse()
  }

```

components/Generator.tsx - streaming to browser

```

import { useCompletion } from 'ai/react'

export function Generator({ profile }) {
  const { completion, complete, isLoading } = useCompletion({
    api: '/api/generate',
  })

  const handleGenerate = () => {
    complete('', {
      body: { contentType, userInput, extraDetail, profile }
    })
  }

  return (
    <div>
      <button onClick={handleGenerate} disabled={isLoading}>
        {isLoading ? 'Writing...' : 'Generate Content'}
      </button>
      <div className="output">{completion}</div>
    </div>
  )
}

```

5.3 Database schema (Supabase / PostgreSQL)

Four tables. Keep it simple.

```

-- Table 1: handled by Supabase Auth (auth.users) - no action needed

-- Table 2: business profiles
create table profiles (
  id          uuid primary key default gen_random_uuid(),
  user_id     uuid references auth.users(id) on delete cascade,
  name        text not null,
  industry    text not null,
  what_sells  text not null,
  target      text not null,
  tone        text not null,
  created_at  timestampz default now()
);

-- Table 3: content generation history
create table generations (
  id          uuid primary key default gen_random_uuid(),
  user_id     uuid references auth.users(id) on delete cascade,
  content_type text not null,
  user_input  text not null,
  output      text not null,
  created_at  timestampz default now()
);

-- Table 4: subscriptions
create table subscriptions (
  id          uuid primary key default gen_random_uuid(),
  user_id     uuid references auth.users(id) on delete cascade,
  plan        text not null default 'free',
  paystack_sub_id text,
  status      text not null default 'active',
  gen_count   int default 0,
  period_start timestampz default now(),
  period_end  timestampz
);

```

5.4 API route structure

Endpoint	Purpose
POST /api/generate	Main AI endpoint. Accepts profile + input, streams response back.
GET /api/profile	Fetch logged-in user's business profile.
POST /api/profile	Create or update business profile.
GET /api/history	Fetch user's content generation history.
POST /api/webhooks/paystack	Receives Paystack events to update subscription status.

GET /api/plan	Return current user plan and generation count for the UI.
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Business Model & Monetisation

6.1 Pricing plans

Three plans. Free tier to get users in, then convert to paid. All billed monthly via Paystack.

Free	Starter	Growth	Pro
N0	N5,000	N12,000	N25,000
Forever free	per month	per month	per month
5 generations total	30 generations/month	Unlimited generations	Everything in Growth
All content types	All content types	All content types	3 business profiles
No card required	Full history	Full history	Early access features
No history	Weekly planner	Weekly planner	WhatsApp broadcast (Phase 2)
No weekly planner	Email support	Priority support	Dedicated support

All prices in Nigerian Naira. Offer 20% annual discount to reduce churn.

6.2 Revenue projections

Conservative estimates based on realistic conversion.

Milestone	Paying users	Mix	Monthly revenue
Month 3 (Launch)	20	15 Starter, 5 Growth	N135,000
Month 6	60	35 Starter, 20 Growth, 5 Pro	N540,000
Month 12	150	80 Starter, 50 Growth, 20 Pro	N1,500,000

Month 18	300	150 Starter, 110 Growth, 40 Pro	N3,070,000
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6.3 Cost breakdown per month

Cost item	Estimate
OpenAI API (GPT-4o-mini)	~N1 per generation. 150 users x 50 avg generations = ~N7,500/month
Vercel hosting	Free tier handles ~100k requests/month. Upgrade at N10,000/month when needed.
Supabase	Free tier: 500MB DB + 50,000 auth users. Free for a long time.
Paystack	1.5% per transaction (capped at N2,000). On N1.5M revenue = ~N22,500/month
Domain + misc	~N5,000/month
TOTAL at N1.5M revenue	~N45,000/month in costs = N1,455,000 gross profit (97% margin)

Break-even calculation

Your fixed costs are near zero. Break-even is: cover your time. If you value your time at N100,000/month, you need just 20 Starter customers to break even. Everything above that is profit. At 60 customers you are earning N500,000+/month net.

Go-to-Market Strategy

7.1 Target customers (who to approach first)

Segment	Why they will pay
Instagram fashion sellers	Lagos-based, post products daily, always complaining about content ideas. High volume.
Food/catering businesses	Post menus, promos, delivery offers. WhatsApp broadcast is their main marketing.
Hair & beauty salons	Active on Instagram, want to look professional but can't afford SMM.
Small logistics companies	Need to announce new routes, promo rates, delivery times via WhatsApp.
Freelancers / consultants	Want to position themselves as experts - need LinkedIn + Twitter content.

7.2 How to get your first 20 paying customers

Week 1-2 (before launch)

Post the problem on your Twitter/X: 'Why do Nigerian small businesses struggle to market consistently?'

Share your build progress on Twitter - 'Building an AI tool that writes captions for Nigerian businesses'

DM 10 business owners you know personally. Ask: 'Would you pay N5,000/month for AI that writes your captions?'

Post in Nigerian entrepreneur WhatsApp groups and Facebook groups

Week 3-4 (soft launch)

Give free 30-day access to 5 beta users in exchange for feedback and testimonials

Post a demo video on Twitter and Instagram showing the tool generating content live

Submit to Nigerian tech communities - Indie Hackers Nigeria, Tech Cabal

Write one blog post: 'How Nigerian SMEs can post on Instagram every day without hiring a manager'

Month 2+ (growth)

Ask every happy user to refer one friend - offer 1 free month for each referral

Partner with business coaches and SME consultants who have audiences

Run targeted Instagram ads to Nigerian business owner accounts (N10,000 test budget)

Post 3x/week on Twitter - marketing tips for Nigerian businesses, showing Postly AI solving them

The most important rule

Do not wait until the product is perfect to start selling. Start talking to potential customers before you write a single line of code. Your first 5 customers should come from your personal network. Build the product alongside real users - not in isolation.

Build Timeline

Estimated for one developer working part-time (~4 hours/day). Full-time would cut this in half.

Week	What to build	Goal
Week 1	Project setup, Supabase auth, basic routing, business profile form	<i>User can sign up and fill in their business profile</i>
Week 2	AI generation route (Vercel AI SDK), streaming output to UI, content type selector	<i>User can generate and see streaming AI content</i>
Week 3	Content history, weekly planner, copy button, UI polish	<i>Core product feels complete and usable</i>
Week 4	Paystack integration, subscription plans, free tier limit enforcement	<i>Users can pay and upgrade plans</i>
Week 5	Bug fixes, loading states, error handling, mobile responsiveness	<i>Product is stable and looks professional</i>
Week 6	Landing page, onboarding flow, beta testing, fixes from feedback	<i>Ready to launch publicly</i>

Rule: ship in 6 weeks no matter what

Do not spend 3 months trying to make it perfect. Ship at week 6 with exactly the MVP features. A working product with 3 paying customers teaches you more than 3 more months of building alone. Add Phase 2 features after you have real users telling you what they need.

Success Metrics

These are the numbers you watch every week. They tell you if the business is healthy.

Metric	What it means & target
MRR (monthly recurring revenue)	Total subscription revenue per month. Goal: N500k by month 6.
Paid conversion rate	% of free users who upgrade to paid. Target: 15-25%.
Monthly churn rate	% of paying users who cancel per month. Target: under 5%.
Generations per user/month	How many times users generate content. High = engaged, low = risk of churn.
DAU / MAU ratio	Daily active / monthly active users. Target: above 30%.
Time to first generation	How fast a new user generates their first content. Target: under 5 minutes.
NPS score	Ask users monthly: 'Would you recommend Postly AI?' Target: above 40.

When to know it is working

Green signals - business is healthy

- Users come back 3+ times per week
- You get referrals without asking
- Users complain when you have downtime
- You are getting feature requests (means they care)

Red signals - fix immediately

- Users sign up but never generate content
- Users generate once and never return
- Free users don't convert after 2 weeks
- Churn is above 10% per month

Now go build it.

You have the skills. You have the stack. You have the idea. The only thing left is to start.

Postly AI

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